LearnHub

App Proposal Document

A cross-platform Learning Management System built with Flutter

Week 1 Submission · Flutter LMS Project · 2026

1. Purpose of the App

LearnHub is a cross-platform Learning Management System (LMS) built with Flutter. Its core purpose is to connect learners with structured, skill-building programs in a clean, distraction-free environment — accessible on mobile phones, tablets, and desktop browsers.

The platform solves a common problem: learners often struggle to find, track, and complete online courses in one unified place. LearnHub brings together program discovery, progress tracking, and achievement recognition in a single cohesive app.

Problem Statement

Many learners lose motivation because they lack visibility into their progress and have no centralised place to manage their learning. LearnHub addresses this by providing a personalised dashboard, clear progress indicators, and an achievement system that rewards continued learning.

Solution

- A clean login screen that gets learners into the app without friction

- A home dashboard showing current progress, popular programs, and earned achievements
- A searchable, filterable program catalog for easy discovery
- Detailed program pages with curriculum overview and a one-tap enrolment button
- A responsive UI that works equally well on phones, tablets, and desktop

2. Target Users

User Type	Description	Primary Goals
Learner Primary	Students, working professionals, or anyone upskilling in tech, design, or marketing	Find programs, track progress, earn certificates
Admin Planned	Platform manager who creates content and manages enrolled users	Create/edit programs, view enrolment data, manage users

Learner Persona

Kavya, 24 — Junior Software Developer
Kavya wants to grow her Flutter skills after hours. She needs a mobile-friendly app she can use on her commute, with clear progress tracking so she stays motivated. She values a clean UI that doesn't get in the way of learning.

Admin Persona

Rahul, 35 — Content Manager
Rahul manages the course catalogue. He needs to add new programs, see how many learners are enrolled, and track completion rates. He primarily uses a desktop browser.

3. Key Features

Learner Features (Week 1 — Implemented)

Feature	Screen	Description
Login	Login Screen	Email + password form, show/hide password toggle, forgot password link, sign-up prompt
Dashboard	Home Screen	Personalised greeting, in-progress course card with progress bar, popular programs grid, achievement stats

Program Discovery	Program List	Search bar, horizontal category filter chips (All / Development / Design / Marketing / Data Science), responsive program grid
Program Detail	Program Detail	Hero gradient header, rating, category badge, description, lesson/hour/level chips, "What you'll learn" checklist, Enrol Now button
Responsive Layout	All Screens	Mobile single-column, tablet 2-column, desktop split/wide layouts using custom breakpoint helpers

Planned Features (Future Weeks)

- User registration and profile management
- Real authentication (Firebase Auth)
- In-app lesson player with video support
- Certificate generation on course completion
- Admin dashboard for program and user management
- Push notifications for learning reminders
- Offline support for downloaded lessons

4. User Journey Examples

🎓 **Learner Journey — Discover and Enrol in a Program**

1. **Open app** — The Login screen loads. Kavya enters her email and password, taps *Log In*.
2. **Home Dashboard** — She sees her in-progress "Flutter Development Basics" course at 75%, and a grid of popular programs.
3. **Browse Programs** — She taps the *Explore* nav item, lands on the Program List. She types "Python" in the search bar and taps the *Development* category chip.
4. **View Details** — She taps *Python for Beginners*, which opens the Program Detail screen showing 25 lessons, 15 hours, Beginner level, and a curriculum checklist.
5. **Enrol** — She taps *Enrol Now*. A confirmation snackbar appears: "*Enrolled in program!*". The program now appears in her dashboard.

🛠️ **Admin Journey — Add a New Program** Planned

1. **Admin Login** — Rahul logs in via the admin portal with his credentials.
2. **Program Management** — He navigates to the *Programs* section of the admin dashboard and taps + *Add Program*.
3. **Fill Details** — He fills in the title, description, category, duration, lessons count, and difficulty level in a form.
4. **Publish** — He taps *Publish*. The new program becomes visible to all learners on the Program List screen.
5. **Monitor** — He returns to the admin dashboard to see enrolment numbers update in real time.

5. Low-Fidelity Wireframes

The following wireframes represent the core layout structure of each screen. Actual implementation matches these layouts in Flutter.

① LOGIN SCREEN — Mobile

```
┌──────────────────────────┐
│ [📚] LearnHub             │
│                          │
│ Welcome back             │
│ Log in to continue...    │
│                          │
│ ┌──────────────────────┐ │
│ │ ✉  Email address     │ │
│ └──────────────────────┘ │
│                          │
│ ┌──────────────────────┐ │
│ │ 🔒  Password     👁   │ │
│ └──────────────────────┘ │
│                          │
│          Forgot password?│
│                          │
│ ┌──────────────────────┐ │
│ │        Log In        │ │
│ └──────────────────────┘ │
│                          │
│ ─────── or ───────       │
│                          │
│ Don't have an account?   │
```

② LOGIN SCREEN — Desktop Split

```
┌─────────────────┬──────────────────┐
│ BRAND PANEL     │  LOGIN FORM      │
│ (gradient)      │                  │
│                 │ Welcome back     │
│ [📚]LearnHub     │                  │
│                 │ ┌────────────┐   │
│ Skills that     │ │ ✉ Email     │   │
│ move your       │ └────────────┘   │
│ career          │ ┌────────────┐   │
│ forward.        │ │ 🔒 Password│   │
│                 │ └────────────┘   │
│ Join thousands  │                  │
│ of learners...  │ [  Log In  ]     │
│                 │                  │
│                 │ ── or ──         │
│                 │ Sign up          │
└─────────────────┴──────────────────┘
```

```
│        [Sign up]        │
└─────────────────────────┘
```

③ HOME / DASHBOARD SCREEN

```
┌─────────────────────────┐
│ Dashboard         🔔    │   ← AppBar
├─────────────────────────┤
│ [K]  Hi, Kavya 👋       │
│      Let's continue...  │
├─────────────────────────┤
│  ┌─────────────────────┐│   ← Gradient ca
│  │ IN PROGRESS         ││
│  │ Flutter Dev Basics  ││
│  │ Module 6 of 8       ││
│  │ ████████░░░░ 75%    ││
│  └─────────────────────┘│
├─────────────────────────┤
│ Popular Programs        │
│  ┌─────────┐┌─────────┐ │
│  │🎨 UI/UX ││🐍 Python││
│  │★4.8(230)││★4.7(185)││
│  └─────────┘└─────────┘ │
│  ┌─────────┐┌─────────┐ │
│  │📊 Data Sc││📣 Mktg  ││
│  └─────────┘└─────────┘ │
├─────────────────────────┤
│ Your Achievements       │
│  ┌──────┐┌──────┐┌──────┐│
│  │  12  ││  8   ││  42  ││
│  │Certs ││Course││Hours ││
│  └──────┘└──────┘└──────┘│
├─────────────────────────┤
│🏠Home │🔍Explore│📚My │👤│   ← Bottom N
└─────────────────────────┘
```

④ PROGRAM LISTING SCREEN

```
┌─────────────────────────┐
│ Programs                │   ← AppBar
├─────────────────────────┤
│  ┌──────────────────┐   │
│  │🔍 Search programs │  │   ← Search bar
│  └──────────────────┘   │
│                         │
│ [All] [Dev] [Design]... │   ← Filter chip
│                         │
│  ┌──────────────────┐   │
│  │🎨 UI/UX Design    │  │
│  │  ★ 4.8 (210)    › │  │
│  └──────────────────┘   │
│  ┌──────────────────┐   │
│  │🐍 Python Beginners│  │
│  │  ★ 4.7 (185)    › │  │
│  └──────────────────┘   │
│  ┌──────────────────┐   │
│  │📊 Data Science    │  │
│  │  ★ 4.6 (160)    › │  │
│  └──────────────────┘   │
│  ┌──────────────────┐   │
│  │📣 Digital Mktg    │  │
│  │  ★ 4.5 (140)    › │  │
│  └──────────────────┘   │
└─────────────────────────┘
```

⑤ PROGRAM DETAIL SCREEN

```
┌─────────────────────────┐
│ ← [gradient header]     │   ← SliverAppBa
│                         │
│  Flutter Development    │
│  Complete Guide         │
├─────────────────────────┤
│ ★ 4.8 (230) [Development│   ← Meta row
├─────────────────────────┤
│ Learn Flutter from      │
│ scratch and build       │
```

⑥ NAVIGATION FLOW DIAGRAM

```
[Login Screen]
      │
      │  Tap "Log In"
      ▼
[Home / Dashboard]
      │
  ┌───┴──────────────┐
  │                  │
  │ Tap program card │ Tap "Explore"
```

```
│  beautiful apps...         │
├────────────────────────────┤
│  [📖 25 Lessons]           │   ← Info chips
│  [⏱ 15 Hours]             │
│  [🏆 Beginner Level]       │
├────────────────────────────┤
│  What you'll learn         │
│  ✅ Flutter Basics & Dart  │
│  ✅ Widgets & State Mgmt   │
│  ✅ Navigation & Routing   │
│  ✅ Firebase Integration   │
├────────────────────────────┤
│  ┌──────────────────────┐  │
│  │      Enrol Now       │  │   ← CTA button
│  └──────────────────────┘  │
└────────────────────────────┘
```

```
   ▼                         ▼
[Program Detail]   [Program List]
   │                         │
   │ Enrol Now               │ Tap program card
   ▼                         ▼
[Snackbar: Enrolled!] [Program Detail]
                              │
                              │ Enrol Now
                              ▼
                    [Snackbar: Enrolled!]

Bottom Nav:
  🏠 Home → Dashboard
  🔍 Explore → Program List
  📚 My Learning → (planned)
  👤 Profile → (planned)
```

6. Technology Stack

Layer	Technology	Reason
Framework	Flutter 3.44 (Dart 3.12)	Single codebase for mobile, web & desktop
UI System	Material Design 3	Modern, accessible, customisable components
State Management	StatefulWidget (local)	Sufficient for Week 1 scope; scalable to BLoC/Riverpod later
Navigation	Named Routes (Navigator 1.0)	Simple, clear route mapping
Responsiveness	Custom Responsive utility	Breakpoints at 599px (mobile) and 1023px (tablet)
Backend (planned)	Firebase (Auth + Firestore)	Real-time data, authentication, scalable

7. GitHub Repository

The project is version-controlled on GitHub. The repository contains:

- Complete Flutter project folder structure
- All source code for 4 screens (Login, Home, Program List, Program Detail)

- Custom theme and responsive utility classes
- This README with project vision, objectives, and navigation flow
- At least one commit per development milestone

Repository URL: `github.com/[your-username]/LearnHub` → Replace with your actual GitHub URL before submission.

Commit History (evidence of version control)

Commit	Message	Description
`initial`	Initial Flutter project setup	Base Flutter project with pubspec.yaml and folder structure
`screens`	Add all 4 core screens	Login, Home, Program List, Program Detail with full UI
`theme`	Add AppTheme and Responsive utilities	Design tokens, colors, spacing, breakpoint helpers
`docs`	Add README and App Proposal	Week 1 documentation deliverables

8. Week 1 Learning Outcomes

By completing Week 1, the following outcomes have been achieved:

- ✅ Defined the app purpose, target users, and key feature set
- ✅ Created user journey examples for both Learner and Admin roles
- ✅ Designed low-fidelity wireframes for all 4 core screens
- ✅ Set up a Flutter project with a clean, scalable folder structure
- ✅ Implemented responsive layouts using custom breakpoint utilities
- ✅ Applied Material Design 3 theming with consistent design tokens
- ✅ Established version control with Git and GitHub
- ✅ Wrote a comprehensive README documenting the project vision and navigation flow